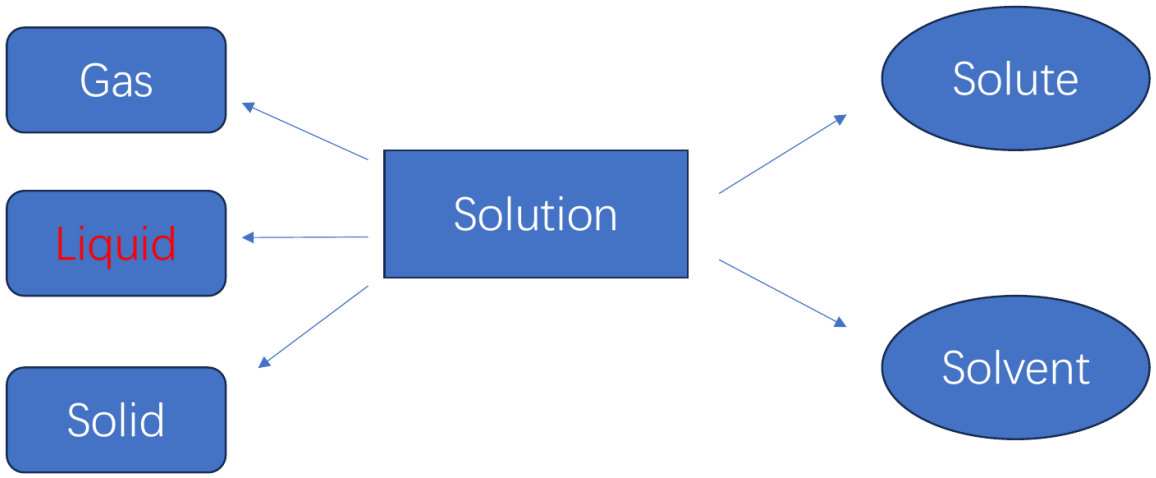
CHEM 1043 Tutorial 3

Solutions and their properties

Types of Solution

A solution is a homogeneous (uniform composition and properties) mixture containing two or more substances in proportions that can be varied. A solution contains solute (less quantity) and solvent (large quantity).



Solution Concentration

A) Percentage Notation:
- Mass Percent (m/m)
- Volume Percent (v/v)
- Mass/Volume Percent (m/v)

B) Parts-per Notation:
- ppm: parts per million (µg/g, mg/L) Million = 10^6
- ppb: parts per billion (ng/g, µg/L) Billion = 10^9
- ppt: parts per trillion (pg/g, ng/L) Trillion = 10^{12}

C) Mole-based Notation:
- Mole Fraction $\chi_i = \dfrac{\text{Amount of component } i \text{ (in moles)}}{\text{Total amount of all components (in moles)}}$
- Mole Percent= Mole fraction x 100%

D) Molarity and Molality:
- Molarity $\text{Molarity } (M) = \dfrac{\text{Amount of solute (in moles)}}{\text{Volume of solution (in liters)}}$ Temperature dependent
- Molality $\text{Molality } (m) = \dfrac{\text{Amount of solute (in moles)}}{\text{Mass of solvent (in kilograms)}}$ Temperature independent

Intermolecular Forces and Solution Process



Why some substances mix to form solutions and others do not?

Oil and water



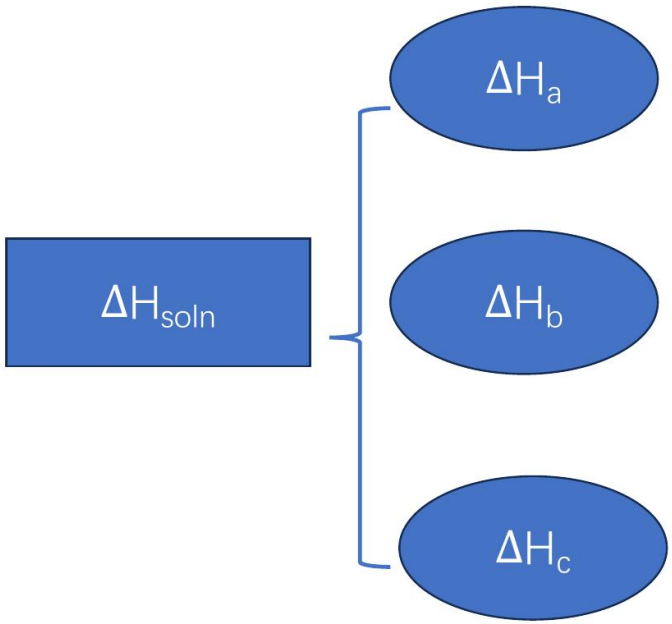


Water and ethanol

In the formation of solutions (solution process), heat change is always involved as a result of redistribution of intermolecular forces. (ΔHsoln) is used to represent the enthalpy change in the solution process.

Enthalpy of Solution

ΔH_{soln} is composed of three components: ΔH_a, ΔH_b and ΔH_c.



Breaking the solvent–solvent attractions. ($\Delta H > 0$)

Breaking the solute–solute attractions. ($\Delta H > 0$)

Forming the solvent–solute attractions. ($\Delta H < 0$)

$$\Delta H_{soln} = \Delta H_a + \Delta H_b + \Delta H_c.$$

Enthalpy of Solution

ΔH_{soln} can be positive (endothermic) or negative (exothermic), depending on the values of ΔH_a, ΔH_b and ΔH_c.

Scenario 1: $\Delta H_{soln} = 0$ Solvent–solute interactions are of equal strength with solvent–solvent and solute–solute interactions. (Ideal solution)

Scenario 2: $\Delta H_{soln} < 0$ Solvent–solute interactionsare stronger than solvent–solvent and solute–solute interactions. (non-Ideal solution)

Scenario 3: $\Delta H_{soln} > 0$ Solvent–solute interactions are somewhat weaker than solvent–solvent and solute–solute interactions. (non-Ideal solution)

Scenario 4: $\Delta H_{soln} >> 0$ Solvent–solute interactions are much weaker than solvent–solvent and solute–solute interactions. (non-Ideal solution)

Intermolecular interaction

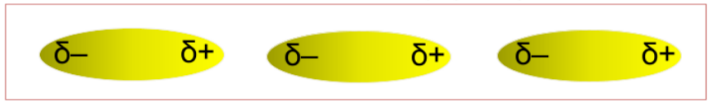
Van der Waals forces refer to a collection of weak attractive forces between groups of atoms or molecules.

Types of van der Waals Forces:

1) Ion – dipole interactions



2) Dipole – dipole interactions



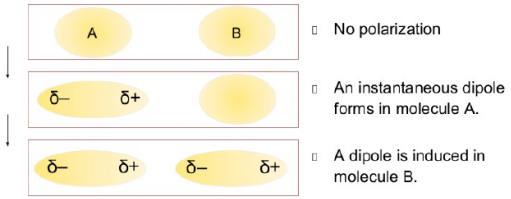
3) Ion – induced dipole interactions



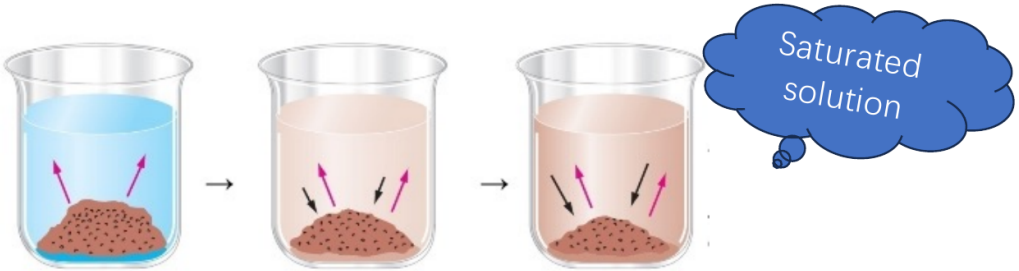
4) Dipole – induced dipole interactions



5) Instantaneous dipole – induced dipole interactions



Solution Formation and Equilibrium



When a solid solute and liquid solvent are mixed:

Stage 1: Only dissolution occurs, no crystallization takes place

Stage 2: Dissolution rate > Crystallization rate

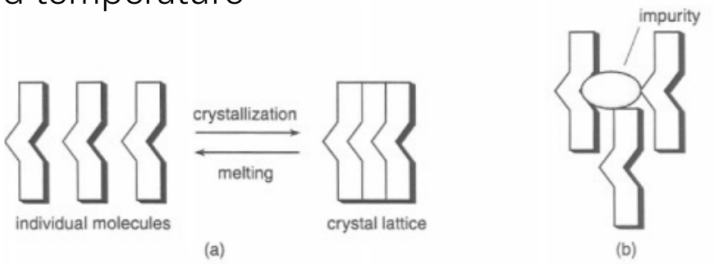
Stage 3: Dissolution rate = Crystallization rate → dynamic equilibrium

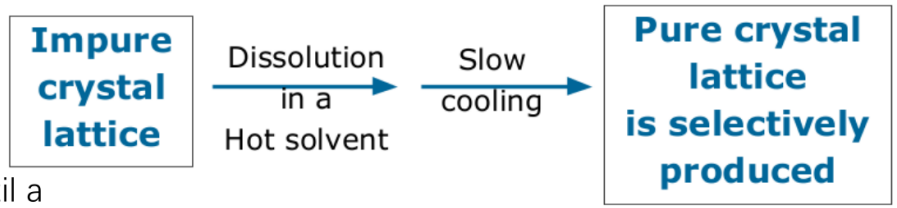
Recrystallization

The increased solubility of a solute in a solvent with increased temperature enables the use of recrystallization for purification.

The Principle of Recrystallization

[desired solute] >> [impurities]





1. Dissolve the solute (desired product) in more soluble solvent.

2. Heat and add the other insoluble solvent dropwise, until a slight cloudiness appears (saturated).

3. Add some of the first solvent again until the cloudiness just
disappears (then add a few more drops).

4. Let the solution cool slowly to initiate the crystallization.

Solubilities of Gases





The effect of temperature on the solubility of gases is generally different from that on solid solutes. In water, the solubilities of most gases [e.g. $N_2(g)$ and $O_2(g)$] decrease with an increase in temperature. The pressure of a gas strongly affects its solubility.

William Henry found that the solubility of a gas increases with increasing pressure.

Henry's law: $C = k \times P_{gas}$

C = solubility of a gas in a particular solvent at a fixed T,

P_{gas} = partial pressure of the gas above the solution

k = proportionality constant.

More about Henry's law

1) In a saturated solution, rate of evaporation of gas molecules from solution = rate of condensation of gas molecules into the solution.

2) Both rates depend on the no. of molecules per unit volume.

3) With increasing pressure on the system,
1. No. of molecules per unit volume in the gaseous state increases (∵ increase in gas pressure).

2. No. of molecules per unit volume in the solution must also increase (∵ increase in concentration).

4) Henry's law applies only to the equilibrium between molecules of a gas and the same molecules in solution.

$$HCl\,(g) \xrightarrow{H_2O} H^+(aq) + Cl^-(aq)$$

Note that Henry's law fails for gases at high pressures; it also fails if the gas ionizes in water or reacts with water.

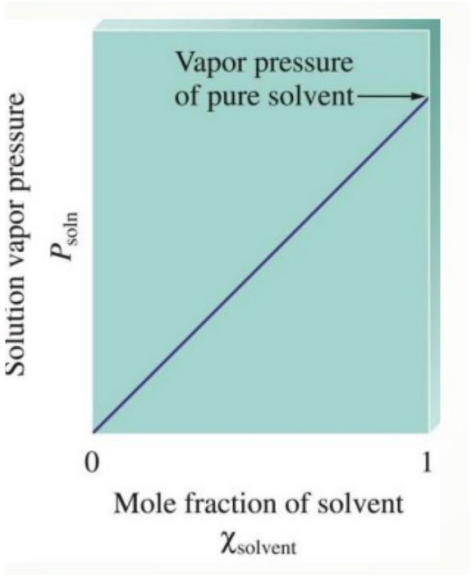
Practice

The 250-mL coke was then opened. Given the atmospheric pressure is 1 atm and volume percent of CO_2 in air is 0.036%, what is the solubility of CO_2 in the coke at 20 °C after exposed to air? K_{CO2} is 0.0365.

Vapour Pressures of Solutions

In the 1880s, F. M. Raoult found that a dissolved solute lowers the vapour pressure of the solvent.

Raoult's law: $P_A = x_A P_A^\circ$



Raoult's law and ideal solution

A dissolved solute lowers the vapour pressure of the solvent.

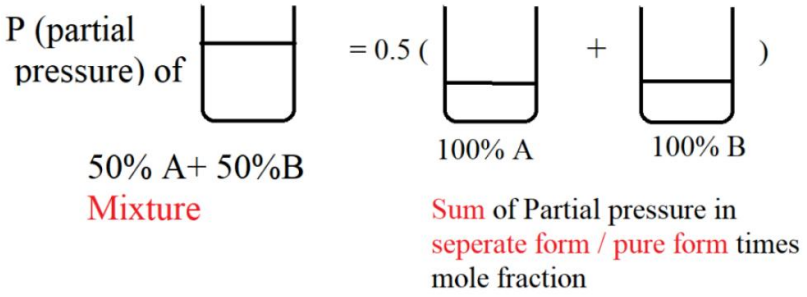
$P_A = x_A P_A^\circ$
X_A is mole fraction of A

P_A°: vapour pressure of the pure solvent at the given temperature

Vapour Pressures of Solutions

Raoult's law and ideal solution

Raoult's law applies only to ideal solutions.
For a 2 components solution mixture:
$PA = x_A P_A^\circ + x_B P_B^\circ$



Practice

The vapor pressure of 50 % (m/v) nitric acid was measured to be 9 mmHg at 25°C and 1 atm. Given that the vapor pressure of pure water and pure nitric acid are 23.8 mmHg and 48 mmHg respectively at 25°C and 1 atm.
Determine whether 50% nitric acid has a positive or negative deviation from Raoult's law based on the given vapor pressure.

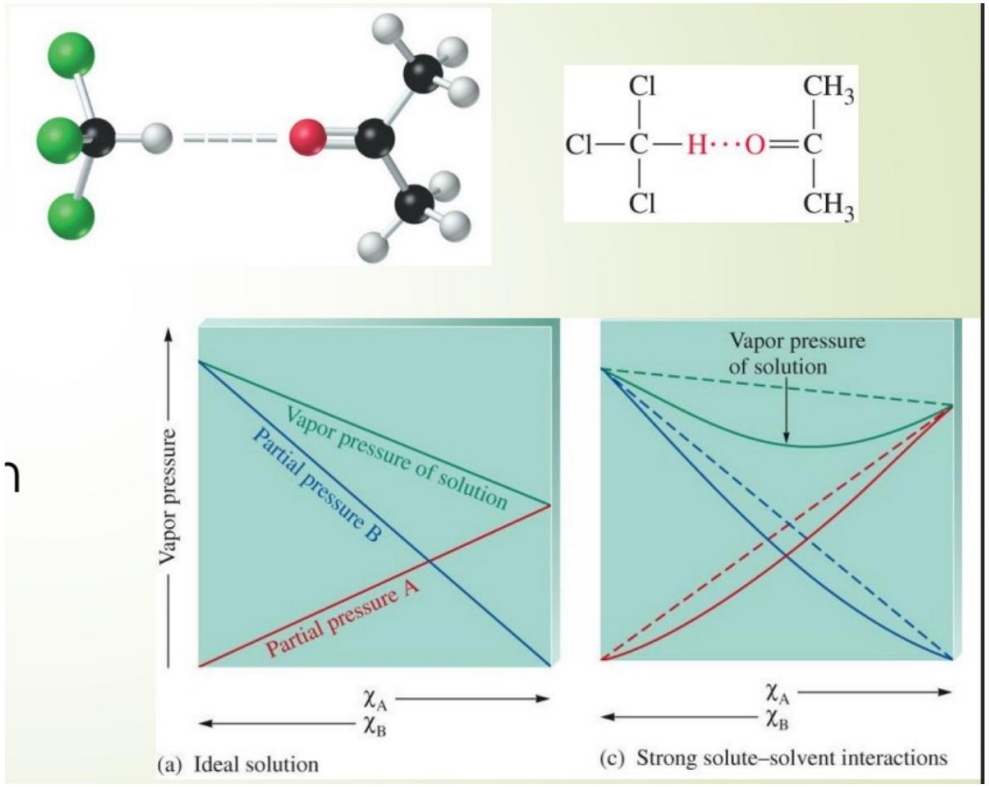
Deviation from Raoult's Law (ideal solution)
Eg: acetone-chloroform

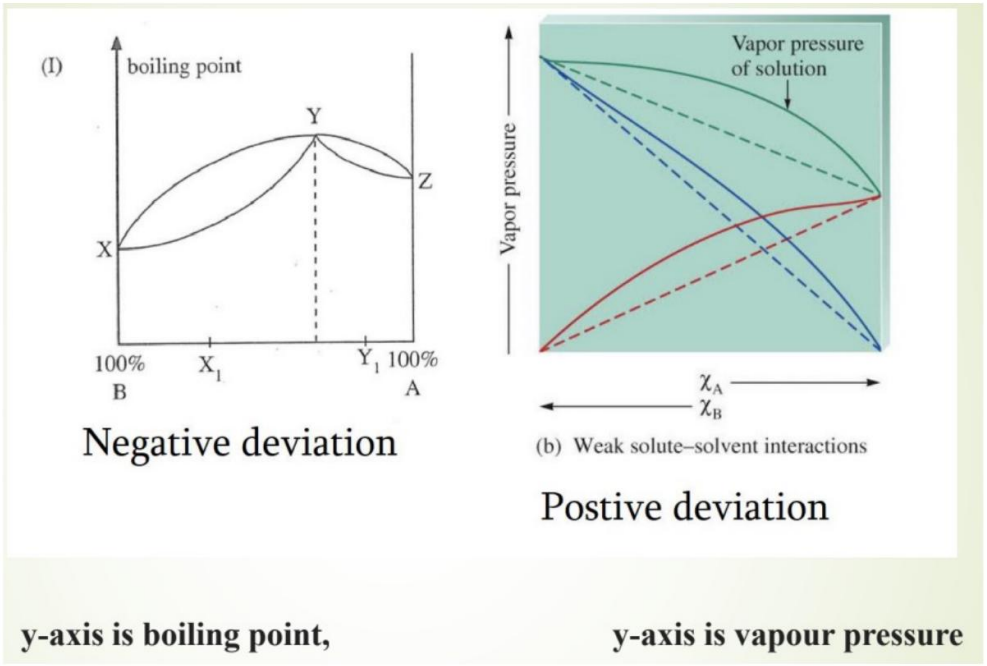
Acetone–chloroform interactions are stronger than interactions in pure acetone and chloroform.

1) More difficult to boil,

2) Lower vapour pressure than Predicted

3) Observed a negative deviation from Raoult's law!

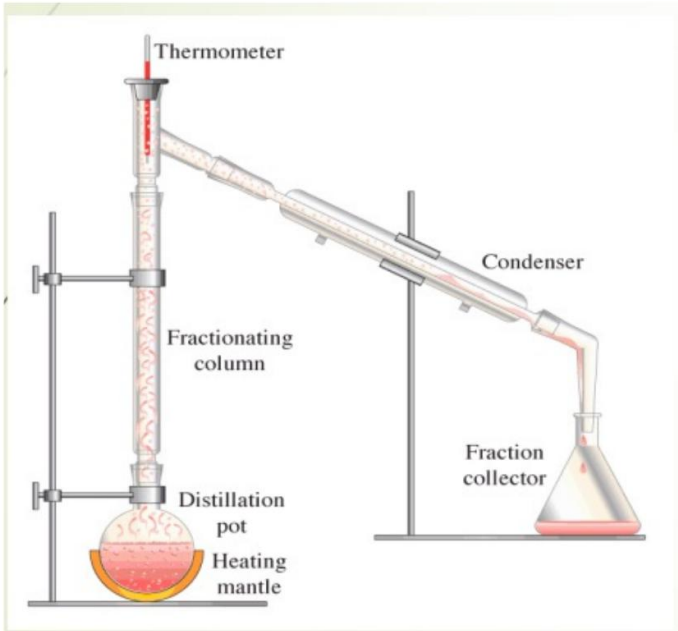


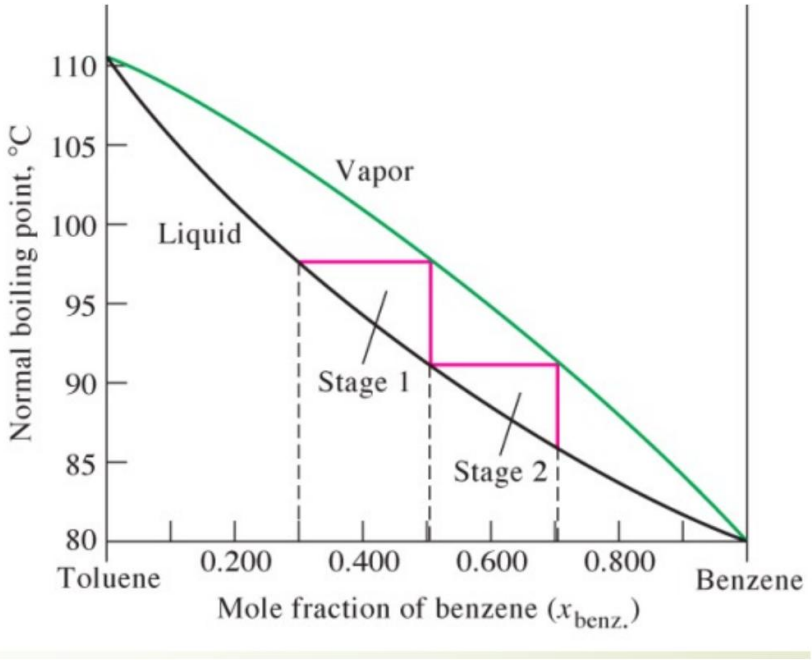


Raoult's law suggests P(total, predicted by Raoult's law) = $P_A x_A + P_b X_b$,
P(total, experimental) - P(total, predicted) is +ve , it is called a positive deviation

Higher boiling point mixture, which means it needs more energy to boil compared to X and Z alone, suggesting that the amount of vapour generated (vapour pressure)

Fractional Distillation

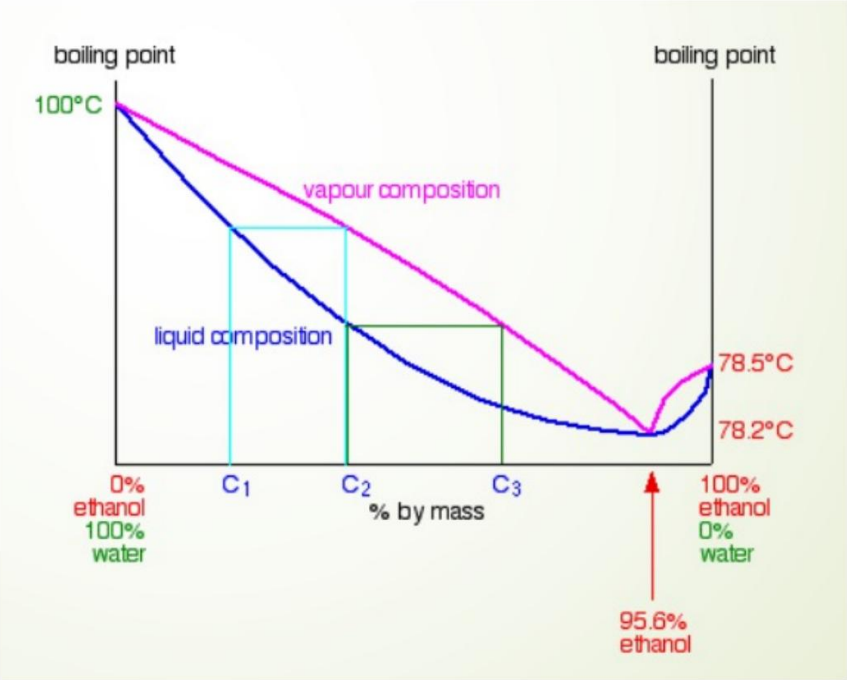




A procedure for separating volatile liquids from one another by boiling.

Purity of stage 2 is higher than stage 1. (higher benzene ratio).

Azeotrope



Boil at a constant temperature and produce a vapour having the same composition as the liquid.

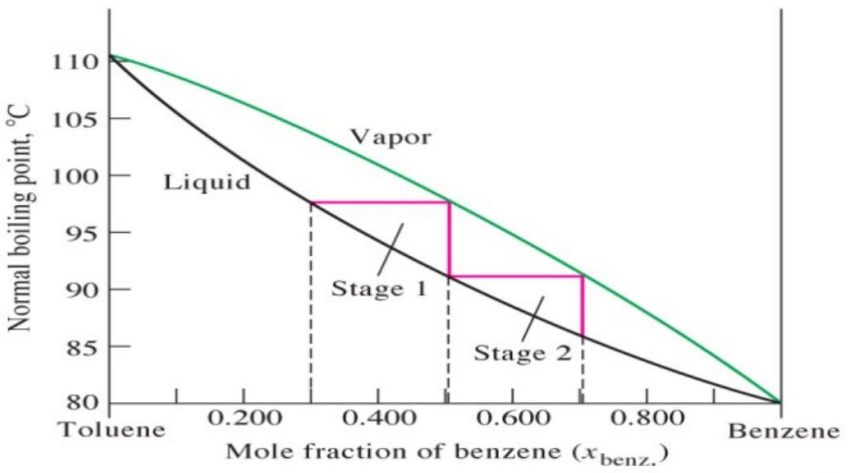
Continuous boiling vs Stage boiling

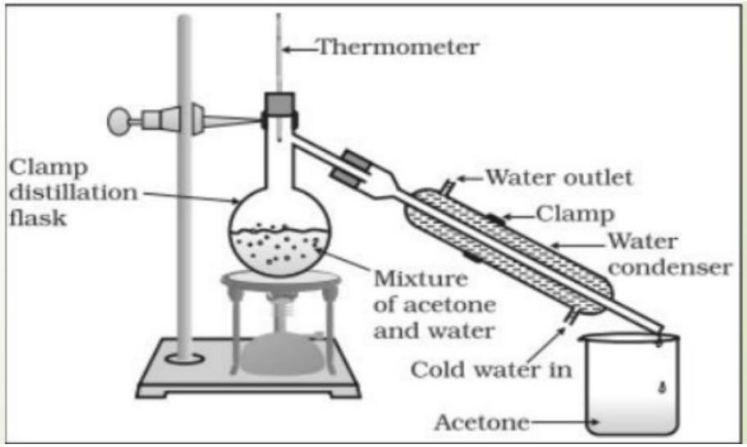
Stage boiling

1) Boil the mixture, collect the distillate,
2) Boil the distillate again in stage two

Use in purification of liquid in liquid mixture

E.g.: increase ethanol content in naturally fermented alcohol



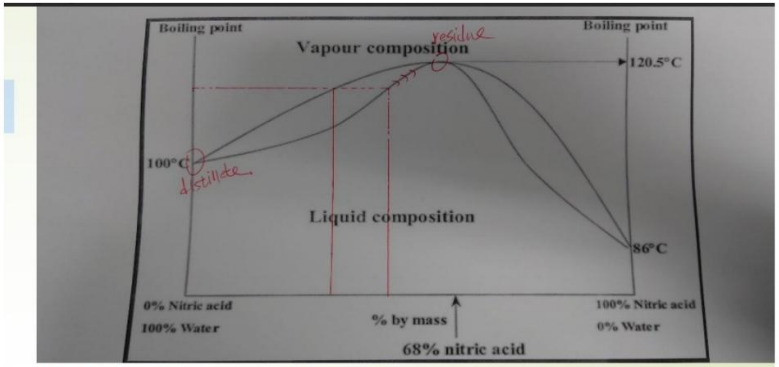


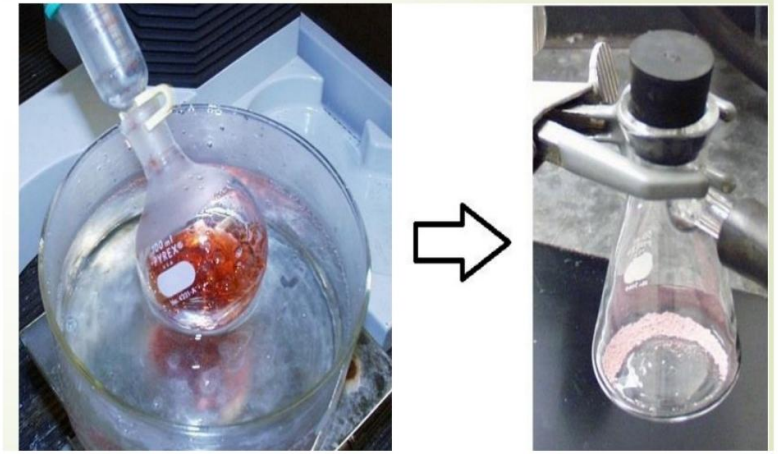
Continuous boiling vs Stage boiling

Continuous boiling

Boil the mixture continuously without isolation of distillate
Only care about final residue and distillate

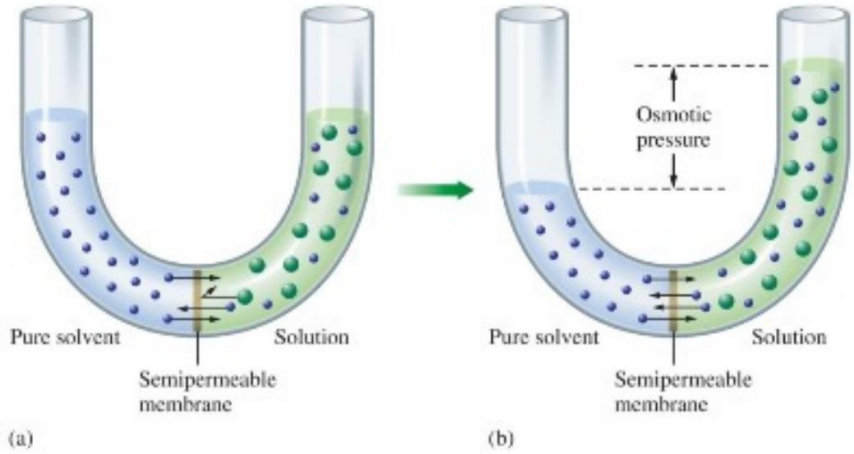
Use in drying up solvent mixture of a chemical reaction
Use in Distillation, to obtain final residue and final distillate





Osmotic Pressure

The concentration of water molecules in the pure water > that in the solution. There is a net flow from the pure water into the solution. This net flow is called osmosis.



$$\pi V = nRT$$

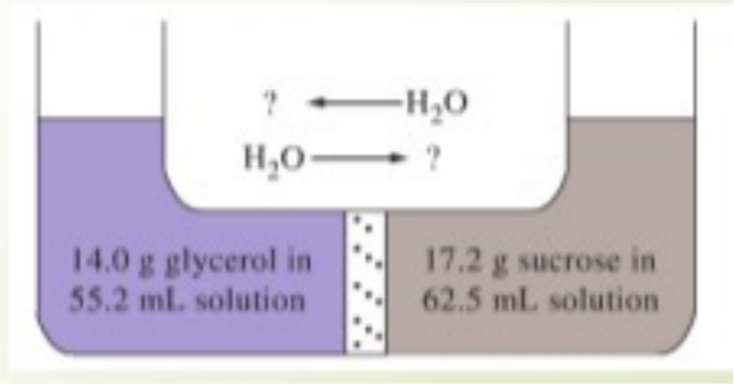
$$\pi = \frac{n}{V}RT = M \times RT$$

M is molarity of solution

where π = osmotic pressure of *dilute* solution,
R = gas constant = 0.08206 L·atm·mol^{-1}·K^{-1},
T = temperature in Kelvin,
n = no. of mole of solute, and
V = volume in litres of solution.

Practice

Two solutions are separated by a semipermeable membrane that permits only the passage of water molecules. In what direction will a net flow of water occur, that is, from left to right or right to left? Glycerol is $OHCH_2CH(OH)CH_2(OH)$; sucrose is $C_{12}H_{22}O_{11}$.



Freezing-point Depression and Boiling-point Elevation of Nonelectrolyte Solutions

The extent to which the freezing point is lowered or the boiling point raised is proportional to the mole fraction of solute (just as is vapour pressure lowering).
In dilute solutions of non-electrolytes (< 1 m), the solute mole fraction is proportional to its molality:

$$\Delta T_f = -K_f \times m$$
$$\Delta T_b = K_b \times m$$

where ΔT_f = the freezing-point depression;
ΔT_b = boiling-point elevation;
m = solute molality; and
K_f and K_b are proportionality constants.

The value of K_f depends on the melting point, enthalpy of fusion and molar mass of the solvent.

The value of K_b depends on the boiling point, enthalpy of vaporization and molar mass of the solvent.

The units of K_f and K_b are $°C\ m^{-1}$.

Their values represent the freezing-point depression and boiling-point elevation for a 1 m solution

Practice

A sample of a certain human hormone weighing 0.546 g was dissolved in 15.0 g benzene, and the freezing-point depression was determined to be 0.240 °C. The freezing-point depression constant for benzene is 5.12 °C·kg/mol. Calculate the molar mass of this hormone.

Colloidal Mixtures

A colloid is a solution that contains particles with one or more of its dimensions (length, width or thickness) in the approximate range of 1-1000 nm.

If all the dimensions are < 1 nm, the particles are of molecular size (solution) ; if all the dimensions are > 1000 nm, the particles are of ordinary/macroscopic size (suspension).



True solution (left): no light is observed.

Colloidal dispersion (right): light is scattered in many directions.

THANK YOU